

# PADHAISHURU

GATE Preparation Platform

## GATE CE

Civil Engineering

### TREND-BASED MOCK PAPER

— 01 —

|                 |            |
|-----------------|------------|
| Duration        | 3 Hours    |
| Total Questions | 65         |
| Maximum Marks   | 100        |
| Sections        | 9 subjects |

#### Difficulty Distribution

Easy: 21 questions (32%)

Moderate: 35 questions (54%)

Difficult: 9 questions (14%)

Unofficial practice material. Not affiliated with or endorsed by GATE/IIT.  
These mock papers are constructed using analysis of historical GATE  
PYQ patterns.  
Historical patterns do not guarantee future examination questions.

# GENERAL INSTRUCTIONS

GATE CE — Trend-Based Mock Paper

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## 1. Examination Pattern

The paper contains 65 questions carrying a total of 100 marks. The duration of the examination is 3 hours.

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## 2. Question Types

Questions are of three types:

- (a) Multiple Choice Questions (MCQ): One correct answer out of four options.
  - (b) Multiple Select Questions (MSQ): One or more correct answers out of four options. Each correct answer carries full marks. There is no negative marking.
  - (c) Numerical Answer Type (NAT): A real number answer to be entered. No options are provided.
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## 3. Marking Scheme

- 1-mark questions: +1 for correct, -0.33 for incorrect (MCQ only)
  - 2-mark questions: +2 for correct, -0.66 for incorrect (MCQ only)
  - 3-mark questions: +3 for correct, -0.66 for incorrect (MCQ only)
  - NAT questions carry no negative marking.
  - MSQ questions carry no negative marking.
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## 4. How to Answer

- For MCQ: Select exactly one option.
  - For MSQ: Select one or more options. Partial credit is NOT awarded.
  - For NAT: Enter the numerical answer in the provided space. Round off as specified.
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## 5. Rough Work

All rough work should be done only in the space provided in this question paper. Do not use additional sheets.

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## 6. Calculator

An on-screen virtual calculator is available for use during the examination. Physical calculators are NOT permitted.

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## 7. Important Notes

- Do not open the seal of the Question Booklet until you are instructed to do so.
  - The Question Booklet contains multiple pages. Verify the page count.
  - Immediately report any discrepancy in the Question Booklet to the invigilator.
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**Q1.** 1M [-0.33]

[General Aptitude]

Volume of sphere r:

- (A)  $\frac{4}{3} \pi r^3$
- (B)  $\frac{4}{3} \pi r^2$
- (C)  $\frac{4}{3} \pi r$
- (D)  $2\pi r^3$

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**Q2.** 1M [no -ve]

[General Aptitude]

Choose similar to 'diligent':

\_\_\_\_\_

Enter your answer here

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**Q3.** 1M [no -ve]

[Engineering Mathematics]

Order of  $\frac{dy}{dx}=y$ :

\_\_\_\_\_

Enter your answer here

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**Q4.** 1M [-0.33]

[Structural Engineering]

Bending moment in UDL simply supported:

- (A)  $wL^2/8$
- (B)  $wL^2/12$
- (C)  $wL/2$
- (D)  $wL^2$

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**Q5.** 1M [-0.33]

[Structural Engineering]

Method of joints gives:

- (A) Forces only
- (B) Moments
- (C) Both
- (D) None

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**Q6.** 1M [-0.33]

[Structural Engineering]

Moment-area method gives:

- (A) Slope
- (B) Deflection
- (C) Both
- (D) None

**Q7.**

1M [-0.33]

*[Geotechnical Engineering]*

Atterberg limits:

- (A) Liquid limit
- (B) Plastic limit
- (C) Shrinkage
- (D) All

**Q8.**

1M [-0.33]

*[Geotechnical Engineering]*

Bearing capacity of footing:

- (A) Depends on soil
- (B) Depends on load
- (C) Both
- (D) None

**Q9.**

1M [-0.33]

*[Water Resources Engineering]*

Rainfall measured by:

- (A) Hygrometer
- (B) Rain gauge
- (C) Anemometer
- (D) Barometer

**Q10.**

1M [-0.33]

*[Water Resources Engineering]*

Darcy-Weisbach for:

- (A) Minor losses
- (B) Major losses
- (C) Both
- (D) None

**Q11.**

1M [-0.33]

*[Water Resources Engineering]*

Manning's equation for:

- (A) Pipe
- (B) Open channel
- (C) Laminar
- (D) Turbulent

**Q12.**

1M [-0.33]

*[Water Resources Engineering]*

Gumbel distribution used for:

- (A) Rainfall
- (B) Flood frequency
- (C) Runoff
- (D) None

**Q13.**

1M [no -ve]

*[Environmental Engineering]*

BOD stands for:

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*Enter your answer here***Q14.**

1M [-0.33]

*[Environmental Engineering]*

Sedimentation removes:

- (A) Dissolved
- (B) Suspended
- (C) Colloids
- (D) Gases

**Q15.**

1M [-0.33]

*[Transportation Engineering]*

PCU stands for:

- (A) A
- (B) B
- (C) C
- (D) D

**Q16.**

1M [-0.33]

*[Transportation Engineering]*

Camber provided for:

- (A) Strength
- (B) Drainage
- (C) Smoothness
- (D) Aesthetics

**Q17.**

1M [no -ve]

*[Surveying & Geomatics]*

Dumpy level used for:

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*Enter your answer here***Q18.**

1M [no -ve]

*[Surveying & Geomatics]*

Theodolite measures:

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*Enter your answer here***Q19.**

1M [-0.33]

*[Construction Materials]*

Water-cement ratio affects:

- (A) Strength only
- (B) Workability only
- (C) Both
- (D) None

**Q20.**

1M [-0.33]

*[Construction Materials]*

Initial setting time OPC:

- (A) 30 min
- (B) 45 min
- (C) 60 min
- (D) 24 hr

**Q21.**

1M [no -ve]

*[Structural Engineering]*

Bending moment in UDL simply supported:

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*Enter your answer here***Q22.**

1M [no -ve]

*[Construction Materials]*

Water-cement ratio affects:

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*Enter your answer here***Q23.**

1M [-0.33]

*[Construction Materials]*

Initial setting time OPC:

- (A) 30 min
- (B) 45 min
- (C) 60 min
- (D) 24 hr

**Q24.**

1M [-0.33]

*[Transportation Engineering]*

Camber provided for:

- (A) Strength
- (B) Drainage
- (C) Smoothness
- (D) Aesthetics

**Q25.**

1M [-0.33]

*[Surveying & Geomatics]*

Theodolite measures:

- (A) Distance
- (B) Angle
- (C) Level
- (D) All

**Q26.**

1M [no -ve]

*[Environmental Engineering]*

Sedimentation removes:

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*Enter your answer here*

Q27.

1M [-0.33]

*[Structural Engineering]*

Method of joints gives:

- (A) Forces only
- (B) Moments
- (C) Both
- (D) None

Q28.

1M [no -ve]

*[Engineering Mathematics]*Order of  $dy/dx=y$ :

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*Enter your answer here*

Q29.

1M [no -ve]

*[Water Resources Engineering]*

Rainfall measured by:

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*Enter your answer here*

Q30.

1M [-0.33]

*[Water Resources Engineering]*

Manning's equation for:

- (A) Pipe
- (B) Open channel
- (C) Laminar
- (D) Turbulent

Q31.

2M [-0.66]

*[Surveying & Geomatics]*

Theodolite measures:

- (A) Distance
- (B) Angle
- (C) Level
- (D) All

Q32.

2M [no -ve]

*[Surveying & Geomatics]*

Dumpy level used for:

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*Enter your answer here*

Q33.

2M [-0.66]

*[Environmental Engineering]*

Sedimentation removes:

- (A) Dissolved
- (B) Suspended
- (C) Colloids
- (D) Gases

**Q34.**

2M [no -ve]

*[Geotechnical Engineering]*

Atterberg limits:

- (A) Liquid limit
- (B) Plastic limit
- (C) Shrinkage
- (D) All

**Q35.**

2M [-0.66]

*[Water Resources Engineering]*

Rainfall measured by:

- (A) Hygrometer
- (B) Rain gauge
- (C) Anemometer
- (D) Barometer

**Q36.**

2M [-0.66]

*[General Aptitude]*

Volume of sphere r:

- (A)  $\frac{4}{3} \pi r^3$
- (B)  $\pi r^3$
- (C)  $\frac{4}{3} \pi r^2$
- (D)  $2\pi r^3$

**Q37.**

2M [-0.66]

*[Environmental Engineering]*

Sedimentation removes:

- (A) Dissolved
- (B) Suspended
- (C) Colloids
- (D) Gases

**Q38.**

2M [no -ve]

*[Surveying & Geomatics]*

Dumpy level used for:

- (A) Angle
- (B) Distance
- (C) Levelling
- (D) All

**Q39.**

2M [no -ve]

*[Geotechnical Engineering]*

Atterberg limits:

- (A) Liquid limit
- (B) Plastic limit
- (C) Shrinkage
- (D) All

**Q40.**

2M [-0.66]

*[Engineering Mathematics]*Order of  $dy/dx=y$ :

- (A) 0
- (B) 1
- (C) 2
- (D) 3

**Q41.**

2M [-0.66]

*[General Aptitude]*

Choose similar to 'diligent':

- (A) Lazy
- (B) Industrious
- (C) Careless
- (D) Ignorant

**Q42.**

2M [-0.66]

*[Geotechnical Engineering]*

Atterberg limits:

- (A) Liquid limit
- (B) Plastic limit
- (C) Shrinkage
- (D) All

**Q43.**

2M [-0.66]

*[Geotechnical Engineering]*

Bearing capacity of footing:

- (A) Depends on soil
- (B) Depends on load
- (C) Both
- (D) None

**Q44.**

2M [-0.66]

*[General Aptitude]*

Volume of sphere r:

- (A)  $\frac{4}{3} \pi r^3$
- (B)  $\pi r^3$
- (C)  $\frac{4}{3} \pi r^2$
- (D)  $2\pi r^3$

**Q45.**

2M [no -ve]

*[Geotechnical Engineering]*

Atterberg limits:

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*Enter your answer here*

**Q46.**

2M [-0.66]

*[Engineering Mathematics]*Order of  $dy/dx=y$ :

- (A) 0
- (B) 1
- (C) 2
- (D) 3

**Q47.**

2M [-0.66]

*[Structural Engineering]*

Bending moment in UDL simply supported:

- (A)  $wL^2/8$
- (B)  $wL^2/12$
- (C)  $wL/2$
- (D)  $wL^2$

**Q48.**

2M [-0.66]

*[Structural Engineering]*

Method of joints gives:

- (A) Forces only
- (B) Moments
- (C) Both
- (D) None

**Q49.**

2M [-0.66]

*[Engineering Mathematics]*Order of  $dy/dx=y$ :

- (A) 0
- (B) 1
- (C) 2
- (D) 3

**Q50.**

2M [-0.66]

*[Water Resources Engineering]*

Gumbel distribution used for:

- (A) Rainfall
- (B) Flood frequency
- (C) Runoff
- (D) None

**Q51.**

2M [-0.66]

*[Transportation Engineering]*

Camber provided for:

- (A) Strength
- (B) Drainage
- (C) Smoothness
- (D) Aesthetics

**Q52.**

2M [no -ve]

*[Structural Engineering]*

Bending moment in UDL simply supported:

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*Enter your answer here***Q53.**

2M [-0.66]

*[Water Resources Engineering]*

Gumbel distribution used for:

- (A) Rainfall
- (B) Flood frequency
- (C) Runoff
- (D) None

**Q54.**

2M [no -ve]

*[Transportation Engineering]*

Camber provided for:

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*Enter your answer here***Q55.**

2M [-0.66]

*[Transportation Engineering]*

PCU stands for:

- (A) A
- (B) B
- (C) C
- (D) D

**Q56.**

2M [-0.66]

*[General Aptitude]*

Choose similar to 'diligent':

- (A) Lazy
- (B) Industrious
- (C) Careless
- (D) Ignorant

**Q57.**

2M [no -ve]

*[Structural Engineering]*

Bending moment in UDL simply supported:

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*Enter your answer here***Q58.**

2M [no -ve]

*[Water Resources Engineering]*

Darcy-Weisbach for:

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*Enter your answer here*

**Q59.**

2M [no -ve]

*[Geotechnical Engineering]*

Atterberg limits:

- (A) Liquid limit
- (B) Plastic limit
- (C) Shrinkage
- (D) All

**Q60.**

2M [no -ve]

*[Environmental Engineering]*

Sedimentation removes:

- (A) Dissolved
- (B) Suspended
- (C) Colloids
- (D) Gases

**Q61.**

2M [no -ve]

*[Environmental Engineering]*

BOD stands for:

- (A) Biological Oxygen Demand
- (B) Basic Oxygen Demand
- (C) Bacterial Oxygen Demand
- (D) None

**Q62.**

2M [-0.66]

*[Water Resources Engineering]*

Manning's equation for:

- (A) Pipe
- (B) Open channel
- (C) Laminar
- (D) Turbulent

**Q63.**

2M [-0.66]

*[Water Resources Engineering]*

Manning's equation for:

- (A) Pipe
- (B) Open channel
- (C) Laminar
- (D) Turbulent

**Q64.**

2M [-0.66]

*[Construction Materials]*

Water-cement ratio affects:

- (A) Strength only
- (B) Workability only
- (C) Both
- (D) None

Q65.

2M [-0.66]

*[General Aptitude]*

Volume of sphere r:

- (A)  $\frac{4}{3} \pi r^3$
- (B)  $\frac{4}{3} \pi r^2$
- (C)  $\frac{4}{3} \pi r$
- (D)  $2\pi r$